

Capstone Project - Digital Marketing

The requirement is to predict the conversion rate (0 or 1) based on several parameters

As a data scientist, I need to develop a model to predict the conversion rate

1.) Identify your problem statement

There are three stages in model prediction

1. Machine learning
2. Supervised learning
3. Classification

2.) Basic Info about dataset

No of rows: 8000 No of columns: 20

3.) Preprocessing methods

- Converted categorical data into numerical data
- Created binary columns to avoid redundancy
- Removed the unnecessary columns

4.) Feature Selection

Applied feature selection like chi-square, Anova f-test, RFE, Wrapper, Random forest, Lasso to get the subset of columns that can be helpful in prediction

5. Model Creation

Based on the features listed, Classification models like logistic regression, KNN, Naive Bayes, Random forest, Decision tree are created

Classification model prediction values

Logistic Regression

CV AUC : 0.8133

CV Acc : 0.8805

CV F1 : 0.9352

Test AUC: 0.8005 | Test Acc: 0.8756

Decision Tree

CV AUC : 0.6842

CV Acc : 0.8611
CV F1 : 0.9206
Test AUC: 0.6419 | Test Acc: 0.8438

Random Forest

CV AUC : 0.9092
CV Acc : 0.8866
CV F1 : 0.9378
Test AUC: 0.9076 | Test Acc: 0.8900

Gradient Boosting

CV AUC : 0.9158
CV Acc : 0.8905
CV F1 : 0.9392
Test AUC: 0.9162 | Test Acc: 0.8881

XGBoost

CV AUC : 0.9005
CV Acc : 0.8873
CV F1 : 0.9369
Test AUC: 0.9014 | Test Acc: 0.8881

AdaBoost

CV AUC : 0.9107
CV Acc : 0.8945
CV F1 : 0.9419
Test AUC: 0.9042 | Test Acc: 0.8850

SVM (RBF)

CV AUC : 0.8484
CV Acc : 0.8869
CV F1 : 0.9392
Test AUC: 0.8286 | Test Acc: 0.8781

K-Nearest Neighbors

CV AUC : 0.7748
CV Acc : 0.8817
CV F1 : 0.9358
Test AUC: 0.7948 | Test Acc: 0.8806

Based on the above models, Random is the best model to evaluate the conversion rate as the accuracy is 0.9076 & ROC value is 0.8900

